

# IAD-Risk: Risk-Aware Identity-Agenda Disentanglement for Scholarly Work Deduplication

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## Abstract

In scholarly data integration, work deduplication must decide whether two records describe the same work, not merely the same research agenda. Identity-agenda confusion arises when semantic relatedness is converted into merge evidence: scientific representation models can rank distinct papers from the same task, benchmark, or citation community as close, yet a single match score does not reveal whether the evidence reflects identity or agenda relatedness. The central insight is that agenda relatedness should become false-merge risk unless independent identity evidence is strong. IAD-Risk implements this insight with stratified data provenance, separate identity and agenda relation heads, agenda-non-identity hard negatives, and a conservative gate before clustering. We evaluate it with IAD-Bench, which separates DeepMatcher gold labels, SciRepEval/SciDocs-style proxy labels, and OpenAlex/OpenCitations silver hard negatives. In the Open-v2 evidence snapshot, result rows are scope-bounded mechanism evidence, not a same-scope comparative ranking. The RoBERTa pair-classifier row remains a strong supervised comparator, not broad-superiority evidence. The single-space scientific representation baselines show HNFMR 0.790–0.999 on the full pair scope; transformer-backed IAD-Risk rows report same-work F1=0.980 and zero observed HNFMR in the held-out hard-negative stratum, with ordinary FMR still reported separately as 0.001. The pair-level conclusion is bounded: identity-agenda disentanglement can reduce risky automatic merges, while cluster artifacts, review-capacity records, external artifacts, and broader validation are required for cluster-level quality, workload reduction, and broad method-ranking claims.

## 1 Introduction

Scholarly data integration and knowledge engineering systems rely on entity matching to decide whether multiple records describe the same scientific work. Digital libraries, literature review tools, citation graphs, recommender systems, and evidence synthesis pipelines all require stable work identities because a false merge can combine evidence from different papers and distort downstream analysis.

This paper focuses on scholarly work deduplication: given two scholarly records with titles, abstracts, authors, venues, identifiers, topics, and citations, the system must decide whether an automatic merge is safe. The output is therefore a selective merge decision, where a pair may be merged, blocked, or deferred. The operating target is not pair classification alone, but same-work utility under explicit hard-negative false-merge control.

The central technical challenge is that agenda evidence can resemble identity evidence. Two papers can share a task, dataset, method family, citation neighborhood, or OpenAlex topic while making different contributions, and a single match score has no explicit place to represent this same-agenda non-identity risk.

Scientific representation models such as SPECTER and SciNCL are valuable because they bring related papers close in embedding space [3, 12]; that behavior supports retrieval and candidate

generation, but it is not sufficient for conservative work identity. A threshold over semantic closeness can select papers that should be reviewed or blocked rather than merged. This failure is particularly important in scholarly settings because false merges are harder to repair than missed duplicates: once distinct papers are clustered as one work, citation counts, recommendation signals, and review evidence can be contaminated. Merge safety therefore requires explicit attention to hard-negative cases where two records are related but non-identical.

We formulate this failure mode as identity-agenda confusion. Identity refers to whether two records denote the same scholarly work. Agenda refers to whether two records discuss the same research problem, benchmark, method family, or citation community.

Existing entity matching methods [5, 8, 11] typically optimize pair classification, while scholarly document representation methods optimize semantic relatedness. Neither objective alone makes the desired decision boundary explicit: same-agenda non-identities should increase false-merge risk rather than merge confidence. The paper therefore treats identity-agenda confusion as a merge-safety failure in scholarly deduplication, not as an ordinary semantic-similarity error.

### 1.1 Motivating Failure Case

A simple failure case makes the distinction concrete. Consider two scholarly records that share the same task or benchmark and use similar terminology, but report a different contribution, model variant, or experimental claim. A scientific representation model may assign high semantic similarity because the records are close in topic space.

For work deduplication, however, the safe decision is to avoid an unsafe automatic merge: without DOI agreement, version evidence, or other same-work identity evidence, the pair should be deferred or blocked rather than merged. The point is that agenda relatedness is not identity evidence; it is a risk condition that should be handled explicitly by HNFMR-style evaluation.

Table 1: Motivating failure case for identity-agenda confusion. The example is an illustrative failure case, not a prevalence estimate.

| Observed pair signal                                          | Agenda interpretation                                             | Merge-safety interpretation                                                                    |
|---------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Same task or benchmark with overlapping technical terms.      | The records are related and may receive high semantic similarity. | Relatedness alone is insufficient for automatic same-work identity.                            |
| Different contribution, model variant, or experimental claim. | The papers can still belong to one research agenda.               | The pair requires same-work identity evidence before merging.                                  |
| No shared DOI, version marker, or clear publication lineage.  | A relatedness model may still rank the pair highly.               | Treating the pair as an automatic merge is an unsafe automatic merge and contributes to HNFMR. |

We propose IAD-Risk, a risk-aware identity-agenda disentanglement framework for scholarly work deduplication. IAD-Risk learns separate relation signals for same-work identity, same-agenda relatedness, and agenda-non-identity hard negatives, then derives a false-merge risk score from these signals. A pair is eligible for automatic merging only when identity evidence is sufficiently strong and agenda-non-identity risk remains below a merge threshold. This selective decision rule turns deduplication from a high-recall similarity problem into a selective safety problem with explicit rejection thresholds.

The paper makes three bounded contributions that align with the failure mode, benchmark contract, and merge mechanism. First, it defines identity-agenda confusion as a scholarly dedu-

plication failure in which agenda relatedness should be measured as false-merge risk rather than treated as merge confidence. Second, it introduces IAD-Bench as a provenance-aware pair contract that keeps gold identity labels, proxy agenda evidence, silver hard negatives, and human-review targets separate. Third, it presents IAD-Risk as a risk-aware merge mechanism that uses separate identity, agenda, and agenda-non-identity signals before applying a conservative merge gate. The Open-v2 evidence snapshot evaluates these contributions under explicit row-scope boundaries; clustering behavior, broad statistical ranking, and artifact-release audit claims remain outside the primary evidence unless the corresponding external artifacts are released.

The current evidence therefore supports a targeted pair-level conclusion: identity-agenda disentanglement can reduce risky automatic merges under the reported stratified evaluation, but it does not support a broad method-ranking claim or a claim that silver labels replace human gold.

Table 2: Contribution-evidence summary. The table states what each contribution is, which evidence supports it, and which stronger claim remains outside the current manuscript.

| Contribution                                                         | Supporting evidence in this paper                                                                                                                                                                                 | Claim boundary                                                                                  |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| Identity-agenda confusion as a scholarly deduplication failure mode. | Problem formulation, hard-negative false-merge rate, and Open-v2 hard-negative evidence.                                                                                                                          | Identifies a concrete safety failure mode, not a prevalence estimate for all scholarly corpora. |
| IAD-Bench as a provenance-aware pair contract.                       | Gold, proxy, silver, and manual-validation layers with label source, label strength, provenance, split, and hard-negative fields.                                                                                 | Defines an evaluation contract; proxy and silver labels are not human identity gold.            |
| IAD-Risk as a risk-aware merge mechanism.                            | Open-v2 evidence snapshot with single-space baseline HNFMR 0.790–0.999 and held-out IAD-Risk same-work F1=0.980 with zero observed HNFMR in the hard-negative stratum; ordinary FMR=0.001 is reported separately. | Supports targeted false-merge suppression, not a broad method-ranking claim.                    |

## 2 Related Work

Entity resolution research defines the broader pipeline in which scholarly deduplication is situated. Classical record linkage formalizes the decision of whether two records refer to the same entity [5], while modern surveys describe end-to-end entity resolution as a workflow involving blocking, matching, and clustering under scale and heterogeneity constraints [2]. System work such as Magellan further emphasizes that practical entity matching requires sampling, labeling, debugging, workflow management, and production execution rather than only a matching function [7]. IAD-Risk follows this pipeline view, but focuses on a specific safety gap in scholarly data: a pair can be highly related in topic or citation context while still being an unsafe merge. From a data-integration perspective, the relevant gap is how the pipeline records provenance, label strength, and relation evidence roles before a merge policy treats relatedness as positive, negative, or deferred evidence.

Neural entity matching methods provide the closest algorithmic baselines for learning pair-level match decisions. DeepER represents tuples with neural distributed representations and supports blocking over learned tuple vectors [4]. DeepMatcher explores design choices for neural entity matching models [11], and Ditto shows that transformer-based matching can benefit from pretraining, fine-tuning, and data augmentation [8]. RoBERTa-style pair encoders [9] are therefore natural baselines for this paper. The distinction is that these methods usually optimize a same-entity clas-

sification objective, whereas IAD-Risk adds explicit same-agenda and agenda-non-identity heads so that semantic relatedness can become blocking evidence rather than merge evidence.

Scientific document representation methods are strong baselines because they intentionally capture citation, topical, and semantic relatedness. SPECTER uses citation-informed supervision to learn document-level scientific representations and introduced SciDocs as a document-level evaluation benchmark [3]. SciNCL improves scientific representations with neighborhood contrastive learning [12], and SciRepEval broadens scientific representation evaluation across classification, regression, ranking, and search while releasing the SPECTER2 model family [14]. These methods are useful for candidate generation and relatedness scoring, but their optimization target is not conservative work identity. IAD-Risk therefore treats high scientific relatedness as a possible false-merge risk unless identity evidence is also strong.

Scientific language resources and open scholarly metadata make reproducible benchmark construction possible, but they do not remove the need for provenance-aware evaluation. SciBERT shows that domain-specific pretraining on scientific text improves scientific NLP tasks [1], and S2ORC provides metadata, abstracts, references, and structured full text for many open-access papers [10]. OpenAlex provides a large open index of scholarly works and concepts [13], while OpenCitations provides open citation data [6]. IAD-Bench uses these resources to construct same-work, same-agenda, and agenda-non-identity evidence, but separates gold, proxy, and silver labels so that citation or topic proximity is never presented as human identity gold.

The closest-work comparison spans End-to-end entity resolution systems, Neural entity matching, Scientific document representations, and Open scholarly metadata benchmarks; the complete positioning matrix is reported in the supplementary material. The comparison serves as a novelty audit rather than another result table: it asks what each line optimizes, why that objective can be unsafe for scholarly work identity, and how IAD-Risk assigns relatedness to false-merge risk gates with gold, proxy, and silver strata for merge-safety framing.

Across these lines, the decisive difference is the decision semantics assigned to relatedness. In end-to-end entity resolution, relatedness often supports blocking or candidate generation; in neural matching, it contributes to a match probability; in scientific representation learning, it defines an embedding neighborhood; and in open metadata benchmarks, it can become a scalable proxy signal. These uses are valuable, but none by itself states whether relatedness is identity evidence, agenda evidence, or agenda-non-identity stress evidence. IAD-Risk makes the narrower claim that relatedness should be interpreted only after the pair schema records that evidence role and before the operating point is fixed: a high relatedness score can become positive, negative, or deferred evidence, not as a direct merge decision. This decision-semantics boundary connects IAD-Bench to HNFMR and separates the contribution from a new encoder or a new threshold over existing embeddings while preserving the gold, proxy, and silver evidence strata.

This positioning narrows the novelty claim to four auditable differences. The paper treats identity-agenda confusion as a false-merge safety failure rather than as generic semantic relatedness, evaluates that failure with HNFMR so agenda-level hard negatives are not hidden inside aggregate pair accuracy, records label strength and provenance so gold identity labels, proxy agenda evidence, and silver stress evidence remain separate, and turns agenda relatedness into risk evidence through explicit relation heads and a merge gate. The boundary remains conservative: IAD-Risk is not a replacement for end-to-end entity resolution workflows and is not a comparative ranking over all neural matching methods; the reported comparisons test the merge-safety framing under the available Open-v2 evidence snapshot. IAD-Bench does not claim that OpenAlex/OpenCitations silver evidence is human gold, but uses those rows as provenance-marked stress evidence for agenda-level false-merge risk.

### 3 Problem Formulation

Let  $d_i$  and  $d_j$  be two scholarly records. The target identity label  $y_{ij}^I$  indicates whether the records describe the same scholarly work. The agenda label  $y_{ij}^A$  indicates whether the records discuss a related research agenda. A hard negative is a pair with high agenda evidence but negative identity evidence:

$$y_{ij}^A = 1, \quad y_{ij}^I = 0.$$

This pair should not be merged even if the documents are semantically close.

We evaluate merge safety with hard-negative false-merge rate:

$$\text{HNFMR} = \frac{\#\{(i, j) : \hat{m}_{ij} = 1, y_{ij}^A = 1, y_{ij}^I = 0\}}{\#\{(i, j) : y_{ij}^A = 1, y_{ij}^I = 0\}},$$

where  $\hat{m}_{ij}$  is the automatic merge decision. The goal is not only to maximize same-work F1, but also to control HNFMR under explicit evidence provenance.

#### 3.1 Notation and Relation Semantics

Table 3 summarizes the notation used throughout the paper. The key distinction is that identity variables describe whether two records denote the same work, while agenda variables describe topical or citation-level relatedness. This distinction prevents a high relatedness score from being interpreted as direct merge evidence.

Table 3: Notation and relation semantics. The paper uses separate symbols for identity, agenda relatedness, agenda-non-identity risk, and merge decisions.

| Symbol or field     | Meaning                                             | Role in evaluation                                                         |
|---------------------|-----------------------------------------------------|----------------------------------------------------------------------------|
| $d_i, d_j$          | Two scholarly records.                              | Input pair for scoring and merge decisions.                                |
| $y_{ij}^I$          | Identity label: the records describe the same work. | Gold identity evidence when supported by the label source.                 |
| $y_{ij}^A$          | Agenda label: the records share a research agenda.  | Proxy or silver relatedness evidence, not sufficient for merging.          |
| $y_{ij}^{ANI}$      | Agenda-non-identity label.                          | Hard-negative supervision for pairs that are related but should not merge. |
| $p_{\text{work}}$   | Predicted same-work probability.                    | Positive merge evidence.                                                   |
| $p_{\text{agenda}}$ | Predicted agenda-relatedness probability.           | Relatedness evidence that can increase risk when identity is weak.         |
| $p_{\text{ani}}$    | Predicted agenda-non-identity probability.          | Direct risk evidence against automatic merging.                            |
| $p_{\text{risk}}$   | Derived false-merge risk score.                     | Gate for rejecting or deferring risky pairs.                               |
| $\hat{m}_{ij}$      | Automatic merge decision.                           | Determines FMR and HNFMR.                                                  |
| label strength      | Gold, proxy, silver, or manual validation layer.    | Controls how a row can be used in training, evaluation, and claims.        |

### 4 IAD-Bench

IAD-Bench is a benchmark contract for organizing scholarly pair evidence. It is not a claim that every label has the same evidential status. Each pair stores relation labels, expected identity labels,

expected agenda labels, label source, label strength, provenance, split, and hard-negative level.

Table 4: IAD-Bench label layers. Gold, proxy, and silver labels are reported separately to avoid overstating evidence strength.

| Layer             | Source                               | Use in this paper                  |
|-------------------|--------------------------------------|------------------------------------|
| Gold              | DeepMatcher / py_entitymatching      | Same-work identity evaluation      |
| Proxy             | SciRepEval / SciDocs-style proximity | Same-agenda analysis               |
| Silver            | OpenAlex / OpenCitations             | Agenda-non-identity hard negatives |
| Manual validation | Independent human review             | External validation target         |

#### 4.1 Document Schema Contract

IAD-Bench first standardizes each scholarly record before constructing pair evidence. The document contract keeps identity-bearing metadata, agenda-bearing metadata, and source provenance in one record so that downstream pair construction can be audited without redistributing raw third-party files. The required fields are grouped as follows:

- record identity: `document_id` and `source_dataset`;
- text and authorship: `title`, `abstract`, and `authors`;
- bibliographic metadata: `year`, `venue`, `doi`, `arxiv_id`, and `openalex_work_id`;
- agenda context: `topics` and `references`.

Missing values are represented by empty strings, empty arrays, or null values rather than by dropping required fields. The full document schema table is reported in the supplementary material.

#### 4.2 Pair Schema Contract

IAD-Bench uses a pair schema that makes relation semantics and evidence strength auditable. The required fields are grouped as follows:

- pair identity: `pair_id`, `source_document_id`, and `target_document_id`;
- relation targets: `relation_label`, `expected_label`, and `expected_agenda_label`;
- evidence source: `label_source`, `label_strength`, and `label_provenance`;
- evaluation control: `split` and `hard_negative_level`.

The schema separates the binary same-work target from agenda relatedness, records the source and strength of each label, and keeps split and hard-negative metadata visible so that training, threshold selection, held-out reporting, and manual audit cannot be silently mixed. It also identifies agenda-level hard negatives for HNFMR. The full pair schema table is reported in the supplementary material.

The Open-v2 benchmark variant used in the core evidence combines 415 DeepMatcher gold pairs with 10,000 OpenAlex-derived silver hard negatives, producing 10,415 pair records over 1,737 documents. The DeepMatcher component provides same-work gold labels, while the OpenAlex component stresses same-agenda non-identity behavior. This design is intentionally conservative: OpenAlex and OpenCitations are used to expose false-merge risk, not to replace human gold.

Table 5: Open-v2 benchmark composition used in the current evidence snapshot. The table separates identity evidence from hard-negative stress evidence before any metric is reported.

| Evidence stratum                                 | Pair count | Evaluation role                                                                            | Claim boundary                                                                                            |
|--------------------------------------------------|------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| DeepMatcher gold identity                        | 415        | Measures same-work matching ability.                                                       | Supports identity evaluation within the imported gold benchmark pairs.                                    |
| OpenAlex and OpenCitations silver hard negatives | 10,000     | Stresses agenda-level false-merge behavior.                                                | Supports hard-negative safety analysis, not human non-identity gold.                                      |
| Open-v2 combined pair scope                      | 10,415     | Provides the shared pair contract for the reported evidence snapshot over 1,737 documents. | Supports the Open-v2 mechanism demonstration; broader source-heldout claims require additional artifacts. |

## 5 Method

### 5.1 Overview

IAD-Risk converts scholarly deduplication into selective risk-aware matching. Given a candidate pair, the model estimates identity, agenda, and agenda-non-identity signals, then derives a false-merge risk score. The final merge decision is made only when identity evidence exceeds a same-work threshold and risk evidence remains below blocking thresholds. Figure 1 summarizes the workflow. The method section follows the same execution order: it first defines identity and agenda signals, then introduces the relation heads and risk score, then specifies the training and inference trace, merge gate, algorithmic audit trail, operational boundary, and implementation details.

The method input is an IAD-Bench pair row joined to two normalized document records, not an unstructured text pair alone. Predictive inputs are identity and agenda features derived from record content and metadata. Label source, label strength, provenance, split, and hard-negative level remain audit fields for masking, threshold selection, evaluation, and release checks; they are not direct predictors. The method output is a set of relation scores, a false-merge risk score, a merge/block/defer decision, and an audit row that binds thresholds, row scope, denominators, and checksums.

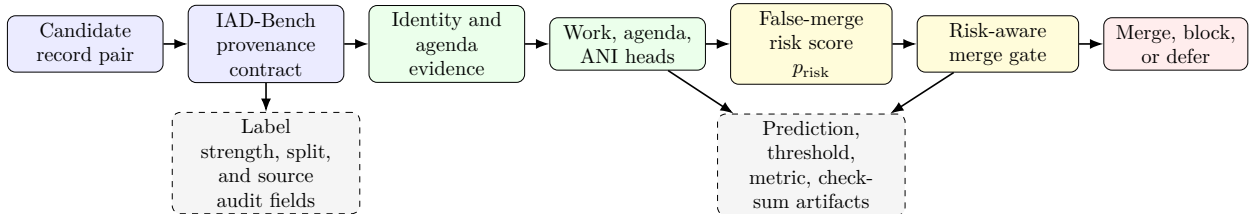


Figure 1: IAD-Risk pipeline. The framework first anchors each candidate pair in the IAD-Bench provenance contract, separates identity evidence from agenda evidence, predicts work, agenda, and ANI heads, derives a false-merge risk score, and then applies a conservative merge gate. Dashed audit nodes mark the schema fields and external artifacts required to reproduce prediction, threshold, metric, and checksum evidence.

## 5.2 Identity and Agenda Signals

IAD-Risk treats identity and agenda signals as separate evidence channels before scoring. Identity evidence includes strong identifiers, near-duplicate titles, author overlap, year and venue agreement, and work-level metadata. Agenda evidence includes shared research topics, similar abstracts, citation neighborhoods, shared references, and scientific representation similarity. This separation addresses feature ambiguity: high embedding similarity may support a same-work hypothesis when identifiers and metadata align, but it may also describe two distinct papers in the same research area. Keeping the channels separate lets the merge gate use relatedness as support, risk evidence, or a reason to defer instead of forcing it into one merge score.

## 5.3 Multi-Task Relation Heads

The model predicts three pair-level relation quantities:  $p_{\text{work}}$ ,  $p_{\text{agenda}}$ , and  $p_{\text{ani}}$ . The same-work head estimates identity. The same-agenda head estimates topical relatedness. The agenda-non-identity head detects pairs that are related but should not be merged. The false-merge risk score is then derived from these heads:

$$p_{\text{risk}} = \max\{p_{\text{ani}}, p_{\text{agenda}}(1 - p_{\text{work}})\}.$$

This construction makes the safety signal high when a pair is explicitly predicted as agenda-non-identity, or when it is agenda-related but lacks strong identity evidence.

## 5.4 Risk Score Design Rationale

$p_{\text{risk}}$  is a conservative upper-envelope risk proxy for automatic merge decisions. The score increases monotonically with agenda-non-identity evidence through  $p_{\text{ani}}$ , and it also increases when agenda evidence is high and identity evidence is weak through  $p_{\text{agenda}}(1 - p_{\text{work}})$ . The max operator keeps either direct ANI evidence or indirect agenda-without-identity evidence sufficient to block automatic merging. This design matches the safety objective of the paper: a pair should defer rather than merge when either source of false-merge risk is high. The product term is not a calibrated probability unless validated against held-out artifacts. Threshold transfer must be rechecked under new source distributions. The full risk score design rationale table is reported in the supplementary material.

## 5.5 Training Objective

The learning objective combines identity, agenda, and agenda-non-identity supervision only through valid supervision channels. Let  $m_{ij}^w$ ,  $m_{ij}^a$ , and  $m_{ij}^n$  indicate whether pair  $(i, j)$  has valid same-work, same-agenda, and agenda-non-identity supervision, respectively. The model minimizes the masked objective

$$\begin{aligned} Z &= \sum_{(i,j)} (m_{ij}^w + m_{ij}^a + m_{ij}^n), \\ \mathcal{L} &= \frac{1}{Z} \sum_{(i,j)} [\lambda_w m_{ij}^w \text{CE}(y_{ij}^I, p_{\text{work}}) \\ &\quad + \lambda_a m_{ij}^a \text{CE}(y_{ij}^A, p_{\text{agenda}}) + \lambda_n m_{ij}^n \text{CE}(y_{ij}^{ANI}, p_{\text{ani}})], \end{aligned}$$

where CE is binary cross-entropy and  $\lambda_w, \lambda_a, \lambda_n$  control the contribution of each supervision channel. Missing labels therefore do not create negative examples for unrelated heads. A mini-batch

with  $Z = 0$  is skipped before the optimizer update, and each reported relation head requires positive mask coverage in the declared training split.

The supervision sources remain label-strength aware. DeepMatcher pairs supervise same-work identity, whereas OpenAlex/OpenCitations silver hard negatives supervise agenda-non-identity risk without being promoted to human gold identity labels. The false-merge risk score is not directly supervised; it is derived after the relation heads are estimated.

## 5.6 Training and Inference Trace

The implementation keeps training, threshold fixation, inference, and metric export in one auditable trace, so relation-head outputs remain connected to the exact decisions used for evaluation. The full training and inference trace table is reported in the supplementary material. This trace is needed for auditability because IAD-Risk is a selective decision system: the same row-level decisions that produce same-work F1 also determine FMR, HNFMR, automatic merge coverage, block rate, and defer rate.

The main execution contract is that schema loading preserves pair IDs and split fields, supervised fitting uses the masked objective without converting gold, proxy, or silver labels into unrelated supervision, threshold fixation records  $\tau_w$ ,  $\tau_n$ ,  $\tau_r$ , threshold source, selection split, and selection metric before held-out reporting, pair scoring writes  $p_{\text{work}}$ ,  $p_{\text{agenda}}$ ,  $p_{\text{ani}}$ , and  $p_{\text{risk}}$ , decision emission records merge, block, or defer with cannot-link status, and metric export binds denominators and checksums to the same prediction artifact. Here  $\tau_n$  denotes the agenda-non-identity risk-head threshold, not an agenda-relatedness threshold.

## 5.7 Risk-Aware Merge Decision

The merge policy is:

$$\hat{m}_{ij} = 1 \quad \text{iff} \quad p_{\text{work}} \geq \tau_w, \quad p_{\text{ani}} < \tau_n, \quad p_{\text{risk}} < \tau_r, \quad \text{cannot\_link}_{ij} = 0.$$

This rule is deliberately selective: uncertain identity or high agenda-non-identity risk prevents automatic merging, while only strong identity evidence under low risk permits it. It can reject or defer high-risk pairs instead of forcing every semantically related pair into a binary match decision. The same decision interface also supports constrained union-find clustering, where cannot-link evidence can block documented conflicts when coverage is available.

Operationally, the three outputs have different meanings. A pair is merged only when all gate conditions pass. A pair is blocked when explicit cannot-link evidence, high agenda-non-identity evidence, or high derived risk makes an automatic merge unsafe under the fixed operating point. A pair is deferred when identity evidence is insufficient for automatic merging but the available evidence does not justify a hard block. This distinction routes false-merge risk into pair-level decisions and audit fields without claiming that every unresolved pair has been fully adjudicated.

## 5.8 Scoring and Merge Algorithm

The executable method follows a fixed scoring and merge order. The full scoring and merge algorithm table is reported in the supplementary material. For each candidate pair, IAD-Risk first loads pair ID, document IDs, label strength, provenance, split, and hard-negative level from the pair schema. It then builds identity, agenda, ANI, and audit fields, including cannot-link feature groups, without using audit metadata as predictors. The relation heads output  $p_{\text{work}}$ ,  $p_{\text{agenda}}$ , and  $p_{\text{ani}}$ ; the derived risk score is  $p_{\text{risk}} = \max\{p_{\text{ani}}, p_{\text{agenda}}(1 - p_{\text{work}})\}$ .

The merge gate combines  $\tau_w$ ,  $\tau_n$ ,  $\tau_r$ , the cannot-link flag, and threshold source to emit merge, block, or defer under a fixed operating point. The implementation then writes prediction and metric audit fields, including decision, row scope, denominators, thresholds, and checksum-bound artifact rows, so that same-work F1, FMR, HNFMR, coverage, and defer-rate audits all refer to the same decision file.

## 5.9 Operational Complexity and Net Benefit

IAD-Risk is appropriate when false merges are more costly than additional review. The additional cost comes from three relation heads and explicit threshold records, rather than from a new end-to-end clustering system. In an operational scholarly index, this cost is justified when a false merge can contaminate citation counts, literature-review evidence, or recommendation signals, and when the system can expose which pairs were merged, blocked, or deferred under the same fixed operating point.

Net benefit also requires workload evidence from the same prediction artifact used for safety metrics: automatic merge coverage must be large enough to reduce reviewer work after accounting for blocked and deferred pairs. Review load should count only workflow-designated review-required outcomes: terminal cannot-link blocks remain safety stops, while unresolved blocks or deferrals contribute to manual handling. The deferral budget and manual-review capacity must therefore be recorded before deployment. Low FMR or HNFMR alone is insufficient for productivity or cost-saving claims. Thresholds are shared operating-point parameters selected on validation evidence, not tuned per pair.

Without workload evidence, IAD-Risk should be read as a conservative safety filter, not a universal replacement for simple deduplication pipelines. The net benefit is strongest in high-stakes scholarly indexes where the cost of one false merge exceeds the cost of reviewing additional deferred pairs, whereas low-risk bulk cleanup tasks may prefer simpler rules. The full operational net-benefit matrix is reported in the supplementary material.

## 5.10 Version and Identifier Boundary

Version and identifier evidence is treated as merge-policy input rather than as a post-hoc cleanup rule. DOI, arXiv, and OpenAlex identifiers provide strong identity cues when they agree, but publication-lineage evidence is still needed when records describe preprint, conference, and journal versions of related work. In this policy, identifier agreement supports merge eligibility, identifier conflict creates cannot-link or defer evidence, and not every related version is automatically the same work. IAD-Risk therefore separates identifier agreement, identifier conflict, and uncertain version boundaries before applying cluster-level merging. The version policy must be declared before cluster-level merging, and manual adjudication is required for ambiguous version boundaries. The full version and identifier boundary table is reported in the supplementary material.

## 5.11 Failure-Control Rationale

IAD-Risk is designed as a failure-control framework rather than another similarity scorer. The full failure-control rationale table is reported in the supplementary material. The main failure-control boundary is that each design choice targets a specific false-merge pathway while leaving unsupported cases explicit.

Topically close papers receive high semantic similarity, so the method separates agenda evidence from identity evidence before the merge gate and requires agenda-level hard negatives for this claim. Silver metadata is treated as if it were human gold only if label strength is ignored, so IAD-Risk

preserves label strength and provenance and keeps silver labels as stress-test evidence rather than exhaustive truth claims. Pairwise errors can contaminate clusters through transitivity, so risk gating and cannot-link evidence are applied before constrained union-find, while cluster-level guarantees still require complete cannot-link coverage.

Thresholds can turn a classifier into an unsafe automatic merger, so work, ANI, and risk thresholds are exposed. Threshold transfer should be rechecked under new source distributions. Proxy labels are over-interpreted when relation support is mixed, so proxy rows remain non-human evidence even when reproducible.

## 5.12 Design Alternatives and Rejected Shortcuts

Several simpler designs are plausible but unsafe for the claim made in this paper. The full design-alternatives table is reported in the supplementary material. The main design boundary is that IAD-Risk uses explicit identity, agenda, and agenda-non-identity signals instead of treating scholarly deduplication as ordinary semantic thresholding or a single pair-classification problem.

Tuning only a representation-similarity threshold is insufficient because high scientific relatedness can still connect different works in the same task or citation community. Relying on one supervised pair classifier remains a relevant shortcut, and RoBERTa remains a strong baseline; however, broad superiority is not claimed because a single score does not expose whether the evidence is identity or agenda relatedness.

Using provenance as a model feature is rejected because source or split identifiers can create leakage, so provenance remains audit metadata. Forcing every candidate into a binary merge decision is rejected because related but under-evidenced pairs should be allowed to defer. Selecting thresholds after test results is rejected because post-hoc threshold choice can hide false-merge risk; threshold stability needs a released grid and checksums.

## 5.13 Implementation Details

The reported transformer variants use frozen scientific document encoders rather than end-to-end fine-tuning. For each candidate pair, the implementation builds three feature groups: identity features combine transformer distances with identifiers, author overlap, venue, and year signals; agenda features combine semantic similarity with topic and citation overlap; agenda-non-identity features add conflict indicators such as different identifiers and year conflict. The three relation heads are trained on the training split, and the merge policy exposes  $\tau_w$ ,  $\tau_n$ , and  $\tau_r$  as configurable thresholds. Unless a validation protocol overrides them, the implementation uses 0.5 for all three thresholds.

## 5.14 Feature and Head Specification

The implementation uses label provenance for supervision control, not as a predictive shortcut. Pair metadata such as label source, label strength, provenance, split, and hard-negative level are copied to prediction outputs for auditing, but they are excluded from the identity, agenda, and risk feature groups used by the relation heads. The full feature and head specification table is reported in the supplementary material.

In the main method narrative, the identity head uses Transformer distances, title similarity, author overlap, and DOI/arXiv/OpenAlex identifier agreement to estimate  $p_{\text{work}}$  with provenance-aware masking. The agenda head uses semantic similarity, topic overlap, and reference Jaccard similarity to estimate relatedness rather than direct merge evidence. The ANI risk head uses topic and reference evidence together with year conflict and different-identifier conflicts to estimate

direct blocking evidence. The risk gate then combines  $p_{\text{work}}$ ,  $p_{\text{agenda}}$ ,  $p_{\text{ani}}$ , thresholds, and cannot-link flags to emit merge, block, or defer decisions, while audit metadata remains traceable but is not a training feature.

## 6 Experiments

### 6.1 Dataset Construction and Evidence Strength

Open-v2 is built by converting public scholarly metadata into a unified pair schema. Deep-Matcher/py\_entitymatching records provide the same-work gold layer, while OpenAlex topics and OpenCitations-style citation signals provide agenda-level silver hard negatives. Each pair keeps its label strength, provenance, split, and hard-negative flag so that a gold identity result is never averaged with a silver stress-test result without being identified as such.

The benchmark therefore has two roles. The gold layer measures ordinary duplicate matching ability, and the silver hard-negative layer measures whether a system over-merges topically close but distinct papers. This separation is important for interpreting the results: a representation baseline can look useful for scientific retrieval while still being unsafe for automatic work identity merging.

### 6.2 Split and Leakage Controls

The evaluation protocol treats split assignment as part of the evidence contract rather than as a preprocessing detail. Each pair carries a split field, label strength, label source, relation label, provenance, and hard-negative level. Training uses only the declared training split, threshold selection uses validation evidence when available, and held-out metrics are reported only after the operating point has been fixed. Metadata fields that identify source, provenance, or split are retained for auditing but are excluded from model features.

The full split and leakage controls table is reported in the supplementary material. The control set includes the train/dev/test split field, an Unordered pair leakage guard, a Document/cluster split-overread guard, a Label-stratum coverage audit, a Source-heldout readiness audit, and a Topic-heldout readiness audit. These controls define what the current evidence supports and what stronger follow-up evidence would require before held-out, source-generalization, or topic-stability claims are made.

These controls reduce two common validity risks. First, the model cannot gain accuracy from explicit provenance or split identifiers because those fields are audit metadata rather than predictive features. Second, a held-out result is interpreted only within the split and coverage regime that generated it. If a split-readiness audit blocks source-heldout or topic-heldout claims, the manuscript keeps the conclusion at the level of random-split or mechanism evidence instead of presenting it as broad generalization. The current split evidence should therefore be read as pair-record held-out mechanism evidence; it does not by itself establish document-disjoint, cluster-disjoint, or unseen-source generalization. Those stronger claims require grouped split manifests, document/cluster overlap reports, per-scope denominators, prediction checksums, threshold logs, and command provenance.

### 6.3 Experimental Questions

The experiments ask four questions, ordered from primary safety evidence to interpretation safeguards. The full evaluation-protocol table is reported in the supplementary material so that the

main manuscript can keep the evidence logic in prose. RQ1 tests whether IAD-Risk preserves same-work matching performance on gold identity pairs with same-work F1. RQ2 tests whether it reduces false merges on silver hard negatives with HNFMR. RQ3 examines whether the observed behavior is consistent with the proposed risk mechanism through FMR and HNFMR. RQ4 tests whether results remain interpretable under gold, proxy, and silver label strata through split metrics. The interpretation boundary is that evidence strength is tied to the label stratum behind each question and to the row scope that generated the metric; gold, proxy, and silver evidence are not mixed into one undifferentiated score.

These questions define the reading order for Table 6. RQ1 is read from same-work F1 on the applicable gold identity scope, RQ2 is read from HNFMR on the silver hard-negative stratum, RQ3 is read from the joint FMR/HNFMR pattern and risk-gate interpretation, and RQ4 is read from the Scope type and label-stratum boundaries rather than from an added performance score. RQ3 is a mechanism-consistency check, not a complete causal ablation claim. RQ4 is an interpretation check over label strata, not a claim that source-heldout or topic-heldout generalization has been fully established. This ordering keeps the result section focused on scope-bounded false-merge evidence rather than on an unsupported same-scope comparative ranking.

## 6.4 Evaluation Metrics

Same-work F1 is computed on pairs whose identity labels come from the gold layer. Ordinary false-merge rate (FMR) is the fraction of non-identity pairs that the system marks for automatic merging. Hard-negative false-merge rate (HNFMR) is the same error measured only on agenda-level hard negatives, where topic or citation evidence is high but identity evidence is negative. Reporting both FMR and HNFMR separates general matching errors from the specific identity-agenda confusion that this paper studies.

## 6.5 Decision-to-Metric Mapping

The reported metrics treat the selective merge decision as a pair-level prediction. In this mapping, automatic merge is the positive decision, while block and defer are non-merge decisions. Deferred same-work pairs reduce recall in same-work F1, so a system cannot improve F1 by deferring positive identity pairs without penalty. FMR and HNFMR count only automatic merges among non-identity rows, while coverage and defer rate must be reported separately before operational automation claims are made. The full decision-to-metric mapping table is reported in the supplementary material.

Operationally, merge is a positive prediction for same-work F1 and is counted as an FMR/HNFMR error when the label is non-identity; block suppresses false merges but does not resolve a positive same-work pair; and defer protects against unsafe automatic decisions but shifts the case to review and lowers recall when the pair is truly same-work.

## 6.6 Metric Formula Boundary

The metric formulas use the selective merge output after thresholding. Let TP denote same-work pairs that are automatically merged, FP denote non-identity pairs that are automatically merged, and FN denote same-work pairs that are blocked or deferred. Same-work F1 is computed as  $2TP/(2TP + FP + FN)$  on the gold identity evaluation rows, where defer and block decisions on true same-work rows enter FN. FMR denominator is all non-identity rows in the evaluated scope and measures unsafe automatic merges rather than manual-review workload, while HNFMR denominator is the agenda-level hard-negative subset and measures identity-agenda false merges

under that stratum. Rows excluded by missing labels are not silently added to denominators; they remain outside the corresponding metric until a valid label stratum is available. The full metric-formula boundary table is reported in the supplementary material.

## 6.7 Threshold Selection and Uncertainty Reporting

Threshold-sensitive systems are evaluated under a fixed selection protocol rather than by choosing the best test threshold after observing results. Merge thresholds are selected on validation evidence when validation rows are available; otherwise, the default implementation threshold is reported as a fixed operating point. The test split is then used only for final metric reporting. This protocol is necessary because a small threshold change can alter FMR and HNFMR more strongly than it alters same-work F1, so the reported table should be read as an operating-point report rather than as a threshold-stability result.

The full threshold and uncertainty reporting protocol is reported in the supplementary material. Stronger uncertainty or mechanism claims require prediction files and resampling logs, ablation outputs for no-risk-gate, no-ANI-head, single-space, no-cannot-link, and post-hoc-threshold variants, and an artifact audit that binds predictions, configs, logs, checksums, and commit identifiers to each reported table. Until those artifacts are released, threshold sensitivity and component causality remain upgrade paths for future evidence rather than conclusions already supported by the current Open-v2 snapshot.

## 6.8 Ablation Acceptance Boundary

Mechanism ablations are accepted only when every variant is evaluated under the same pair scope, split field, metric implementation, and predeclared operating-point rule as the full IAD-Risk row. The required variants are no-risk-gate, no-ANI-head, single-space, no-cannot-link, and post-hoc-threshold; the executable ablation summary must expose these names through a `protocol_variant` field. Each variant must release prediction rows, threshold logs, denominator records for same-work F1, FMR, and HNFMR, and checksum-bound configuration files before it can support a component-causality statement. If any variant changes the evaluated pair universe, label stratum, threshold-selection source, or prediction schema, the result is treated as exploratory diagnostic evidence rather than an accepted ablation for causal interpretation. The post-hoc-threshold row is a diagnostic threshold-sweep control and cannot by itself support a component-causality claim.

## 6.9 Statistical Interpretation Boundary

The reported Open-v2 numbers are point estimates for a fixed evidence snapshot, not statistical superiority estimates. Confidence intervals, significance tests, and model-ranking statements are intentionally withheld unless the released artifact package contains the exact prediction files, resampling logs, random seeds, and checksums used to compute them. In particular, an HNFMR value of 0.000 means that no hard-negative false merge was observed under the reported pair scope and operating point; it should not be read as proof of zero risk under all source distributions.

Zero-observed HNFMR rows require numerator-denominator disclosure before numerical audit. The corresponding artifact row must record hard-negative false-merge numerator = 0, the HNFMR denominator, the hard-negative label stratum, the evaluated split, the threshold source, and the prediction-file checksum. A rounded value of 0.000 without those fields is treated only as a manuscript summary, not as independently auditable evidence.

Confidence-interval claims require `bootstrap_intervals` generated from the same prediction files. Predefined tests, multiplicity handling, input artifacts, and reproducible analysis logs are re-

quired before statistical-significance wording. Same-scope predictions, interval estimates, ablations, and manual-validation slice are required before stronger model-ranking claims. The full statistical interpretation table is reported in the supplementary material.

## 6.10 Operating Point Disclosure

The main result table reports fixed operating points rather than post-hoc best test thresholds. The full operating-point disclosure table is reported in the supplementary material. The main text keeps the operating-point rule as prose: Representation cosine baselines use cosine score thresholds and require a score file, metric summary, and threshold entry; the RoBERTa pair classifier uses a pair match probability threshold and requires a prediction file, metric summary, and model log; and IAD-Risk transformer variants use  $p_{\text{work}}$ ,  $p_{\text{ani}}$ , and  $p_{\text{risk}}$  under the risk gate, with default  $\tau_w = \tau_n = \tau_r = 0.5$  unless the artifact config records an override. Exact threshold values, selected validation sources, command logs, prediction file, model JSON, thresholds, and checksums must be verified from the released artifact manifest before using the table for numerical reproduction.

Default-threshold rows have a narrower interpretation. A row that uses  $\tau_w = \tau_n = \tau_r = 0.5$  must expose `threshold_source=predeclared_default`, a pre-run configuration checksum, a configuration timestamp or run identifier, and evidence that the threshold values were fixed before held-out scoring. Such rows cannot be described as validation-selected, optimized, or threshold-stable unless `threshold_selection_logs` and `threshold_sensitivity_grid` are released from the same prediction file.

## 6.11 Selective Decision Coverage Boundary

IAD-Risk is a selective decision system, so false-merge metrics should be interpreted together with coverage and review-load evidence. A low FMR or HNFMR is not by itself a throughput claim if the operating point blocks or defers many pairs. A result with low FMR or HNFMR but high deferral is a conservative triage result rather than evidence of reduced review cost. The full selective-decision coverage boundary table is reported in the supplementary material.

The main text keeps the operational boundary as a prose rule: automatic merge coverage, block rate, defer rate, review load, and capacity-normalized review load must be computed from the same prediction files and compared with a predeclared manual-review capacity and deferral budget before making operational automation claims. The current manuscript therefore reports false-merge control under fixed operating points, but does not claim throughput reduction and does not claim all-pair automatic resolution.

Let  $N$  be the number of evaluated pairs under a fixed operating point, and let  $M$ ,  $B$ , and  $D$  be the numbers of automatic merge, block, and defer decisions from the same prediction artifact, so  $N = M + B + D$ . The required coverage quantities are

$$\text{AMC} = \frac{M}{N}, \quad \text{BR} = \frac{B}{N}, \quad \text{DR} = \frac{D}{N}.$$

Let  $B_{\text{review}} \leq B$  and  $D_{\text{review}} \leq D$  denote the blocked and deferred pairs that the declared workflow sends to human review. Terminal cannot-link blocks that need no further adjudication remain counted in  $B$  but are not counted in  $R$ . If  $R = B_{\text{review}} + D_{\text{review}}$  and  $C$  denotes the predeclared manual-review capacity for the same batch, the capacity-normalized review load is

$$\text{CNRL} = \frac{R}{C}.$$

These quantities must use the same prediction artifact, threshold configuration, row scope, and denominator record as FMR and HNFMR; otherwise low false-merge rates can be produced by shifting unresolved pairs to manual review. The current result table therefore supports fixed-threshold false-merge control, not an operational throughput or staffing claim.

### 6.12 Pair-to-Cluster Evidence Boundary

The reported pair-level metrics do not by themselves prove cluster-level deduplication quality. Pair decisions can propagate through transitive merge propagation when a deduplication system turns accepted links into connected components. A single unsafe merge can contaminate a larger cluster, while one cannot-link decision can block a transitive path that would otherwise merge distinct works. The full pair-to-cluster evidence boundary table is reported in the supplementary material. The main text keeps the deployment boundary as a prose rule: cluster assignments, cannot-link violations, `cannot_link_audit`, `cluster_metric_summary`, pair-to-cluster trace files, and cluster contamination rate are required before pair-level false-merge control can be interpreted as a cluster-level deployment guarantee. The paper therefore does not claim cluster-level contamination is eliminated.

### 6.13 Threshold Sensitivity Evidence Status

Threshold stability is treated as an audit requirement, not as an unsupported robustness claim. The current Open-v2 table reports fixed operating points and therefore supports the mechanism interpretation under those operating points. The full threshold-sensitivity evidence status table is reported in the supplementary material. The main text keeps the threshold boundary as a prose rule: the current manuscript reports fixed operating points, the threshold grid is not reported as primary evidence, and metric stability requires the same prediction files, predefined threshold ranges, per-threshold F1, FMR, HNFMR, pair counts, random seeds, command logs, a manifest, and checksums. The current result therefore supports fixed-threshold false-merge control, not threshold-stable ranking across all operating points.

### 6.14 Baselines

The main table reports two scientific representation cosine baselines and one supervised pair-classifier baseline: SciNCL cosine, SPECTER2 adapter cosine, and RoBERTa pair classification. These rows are evaluated through the same pair schema so that same-work F1 and hard-negative false-merge rate can be interpreted without mixing label strengths. This schema alignment makes each row auditable, but it does not by itself make rows with different available result scopes a same-scope ranking. Other repository baseline utilities are not used as primary manuscript evidence unless their metric summaries, prediction files, thresholds, and checksums are included in the released artifact package.

### 6.15 Baseline Inclusion Rationale

The baseline set is selected by evidence availability and evaluation relevance rather than by convenience. The complete inclusion matrix is reported in the supplementary material so that the main manuscript states only the decision rule. The prose keeps exact identifier matching and title-normalization rules as important sanity checks; their current boundary is: Excluded from primary result table when only utility code or fixture-level checks are available. Traditional entity-resolution

systems use the artifact rule: Included as primary evidence only when metric summaries, prediction files, threshold records, and checksums are available. Scientific representation baselines are included because SciNCL cosine and SPECTER2 adapter cosine have reported score summaries under Open-v2, and RoBERTa pair classification is included as a supervised comparator while keeping the result below broad ranking claims.

Omitting a baseline family from Table 6 is not a claim that omitted baselines were outperformed. It means the current manuscript does not yet have the artifact-backed evidence needed to make that row auditable. A same-scope baseline matrix is required before broad ranking claims, especially for exact identifier matching, title-normalization rules, and traditional entity-resolution systems.

## 6.16 Baseline Fairness Controls

Baseline comparison is controlled through the pair contract rather than through a broad comparative-ranking claim. All baselines consume the same IAD-Bench pair records for the row scope being reported, and each row uses the same train/dev/test split field when training is required. Threshold-sensitive rows use validation-selected operating points before held-out reporting, and all systems report same-work F1, FMR, and HNFMR from the same metric implementation. Label source, provenance, and split identifiers remain audit fields, not predictive features. These controls support row-level auditability and mechanism interpretation; a stricter comparative ranking still requires same-scope released prediction files, and rows with different available scopes should not be read as a single comparative ranking. The full fairness-control matrix is in the supplementary material.

## 6.17 Implementation and Reproducibility

All reported systems are evaluated through the same pair-level schema, split field, and metric implementation. Threshold-sensitive baselines select merge thresholds on validation evidence and are then evaluated on held-out evidence. The main metrics are same-work F1 on gold identity pairs, ordinary false-merge rate, and HNFMR on agenda-level hard negatives. Rows with different pair counts in Table 6 should therefore be read as evidence for the same mechanism under different available result scopes, not as a single comparative ranking.

The repository separates executable code from large data artifacts: raw third-party data remain outside Git, while processing scripts, fixture tests, schema contracts, and artifact-release rules remain version controlled. The executable data-processing path is part of the paper evidence rather than a separate engineering convenience: `prepare-deepmatcher`, `prepare-scirepeval-proximity`, and `prepare-openalex-weak-labels` convert public source files into normalized document and pair records, and `build-iad-bench` assembles split, provenance, summary, and dataset-card outputs under the IAD-Bench schema. Generated outputs are then checked through manifests and checksums. This organization is intended to make the main evidence reproducible without redistributing restricted or large source files. The supplementary material states the claim-evidence boundary for each major conclusion so that code-level reproducibility, artifact-level reproducibility, and manual-validation requirements are not conflated.

## 6.18 Artifact Reproduction Protocol

Because the repository does not redistribute large third-party data, reproducibility is divided into code-level reproduction and result-level artifact reproduction. The full reproduction-level table is reported in the supplementary material. The main boundary is that L0 code check and L1

fixture rebuild verify executable contracts, command-line discovery, public-release scans, tests, PDF builds, and miniature fixture adapters under `tests/fixtures`; they do not reproduce the Open-v2 numerical table. In practical review terms, L0/L1 checks answer whether the released code path runs, L2 public-source rebuild answers whether independently obtained public files can be converted into the same artifact classes, and L3 result audit answers whether released predictions, thresholds, logs, and manifests reproduce the reported numerical rows. L2 public-source rebuild requires independently obtained public raw files plus `source_input_manifest` and `processing_run_log` so that input provenance, commands, software versions, seeds, output summaries, and checksums can be audited. L3 result audit requires released tables, predictions, logs, manifests, checksums, and commit identifiers before numerical results can be treated as externally auditable.

The key principle is that small fixtures verify processing contracts, while full reported tables require either independently obtained public source files or a released artifact package with manifests and checksums. A reviewer who has only the Git repository can evaluate installation, schema behavior, fixture rebuilding, public-release boundaries, and manuscript construction. A reviewer cannot verify model predictions, threshold choices, or row-level Open-v2 numbers from Git alone without the L2/L3 artifact chain.

## 6.19 Claim-Evidence Boundary for Result Interpretation

The result interpretation follows a conservative claim-evidence boundary. Each reported conclusion must be tied to a label stratum, an output artifact, and a statement of what the evidence does not prove. This boundary is especially important because the benchmark combines gold identity pairs with proxy and silver stress-test evidence rather than treating all rows as one homogeneous test set.

The full claim-evidence boundary table is reported in the supplementary material. The main text keeps the boundary as a prose rule: identity-agenda confusion is supported only as a false-merge pathway; IAD-Risk support is bounded to the reported Open-v2 setting; IAD-Bench is a provenance-aware evaluation contract; and repository-level reproduction does not by itself prove full numerical results without public source files or a released artifact package.

## 6.20 Result Audit Trail

Table 6 should be read together with the claim-evidence boundary stated above and expanded in the supplementary material. Each numerical row must be traceable to three audit objects: a prediction or score file, a metric summary produced from that file, and a checksum or manifest entry that fixes the artifact version. The same audit record must also contain per-row denominator counts for same-work F1, FMR, and HNFMR, the per-row threshold source, the scope label used in the main table, and the selective-decision coverage fields: automatic merge count, block count, defer count, automatic merge coverage, defer rate, and capacity-normalized review load.

Prediction JSONL artifacts must expose system, pair/document IDs, expected labels, label strength, hard-negative level, split identifiers, score or probability fields, threshold source, threshold values where applicable, and merge decisions; threshold logs must expose threshold name, value, selection split, selection metric, selection rule, applied scope, and score field. Public-source rebuild evidence must also include `source_input_manifest` and `processing_run_log`, because these files bind public inputs, commands, code commits, environments, seeds, output paths, and exit status to the derived evaluation artifacts.

The Git repository verifies the schema, fixture rebuild, public-release boundary, manuscript build, and submission package, but it deliberately does not store raw third-party data, large model

outputs, or full prediction files. Therefore the manuscript treats the Open-v2 table as a supported evidence snapshot and reserves full numerical reproduction for the released L3 artifact package.

The complete row-family artifact crosswalk is reported in the supplementary material. It binds `open_v2_main_results`, `iad_bench_split_summary`, `representation_baseline_scores`, `supervised_baseline_predictions`, `iad_risk_predictions`, `threshold_selection_logs`, `bootstrap_intervals`, `ablation_suite`, `manual_validation_slice`, and `threshold_sensitivity_grid` to the row scopes, operating points, optional stronger claims, and checksum-fixed artifact records used for full numerical audit.

## 6.21 Main Open-v2 Results

The Scope type column is part of the result interpretation: full Open-v2 rows and held-out test rows support a mechanism comparison, not a cross-row ranking.

The IAD-Risk rows also use a fixed operating point. Unless an artifact row records a validation-selected override,  $\tau_w$ ,  $\tau_n$ , and  $\tau_r$  are treated as predeclared default thresholds rather than as thresholds optimized on the reported test rows. This convention makes the decision rule auditable, but it is not evidence that the default thresholds are optimal or stable across source mixtures. Threshold-stability or model-ranking claims require released threshold-selection logs, threshold-sensitivity grids, prediction files, and checksums generated from the same row scope.

Table 6: Open-v2 evidence snapshot. Lower false-merge rate is better. Scope type separates full Open-v2 and held-out test rows. The Denom. audit column means that same-work F1, FMR, and HNFMR denominators must be present in the corresponding `open_v2_main_results` artifact row before numerical audit.

| System                  | Scope type    | Pairs | Denom.       | audit | F1 ↑  | FMR ↓ | HNFMFMR ↓ |
|-------------------------|---------------|-------|--------------|-------|-------|-------|-----------|
| SciNCL cosine           | Full Open-v2  | 10415 | Artifact row |       | 0.054 | 0.785 | 0.790     |
| SPECTER2 adapter cosine | Full Open-v2  | 10415 | Artifact row |       | 0.044 | 0.999 | 0.999     |
| RoBERTa pair classifier | Full Open-v2  | 10415 | Artifact row |       | 0.825 | 0.001 | 0.0001    |
| IAD-Risk (SciNCL)       | Held-out test | 1042  | Artifact row |       | 0.980 | 0.001 | 0.000     |
| IAD-Risk (SPECTER2)     | Held-out test | 1042  | Artifact row |       | 0.980 | 0.001 | 0.000     |

Figure 2 visualizes two selected dimensions from Table 6: same-work F1 and HNFMR. The figure is intended as a mechanism-evidence profile rather than as a complete metric replacement for the table: it makes the hard-negative false-merge exposure of single-space representation baselines visible while keeping the full-scope and held-out test rows separated. Ordinary FMR, pair counts, and denominator-audit status remain table-level evidence and should be read directly from Table 6.



Figure 2: Open-v2 mechanism-evidence profile for selected metrics. The bars visualize same-work F1 and HNFMR from Table 6; ordinary FMR, pair counts, and denominator-audit status remain in the table. Full Open-v2 rows and held-out test rows remain scope-labeled, the dashed separator marks that scope transition, and the figure supports the false-merge-control reading rather than a same-scope comparative ranking.

The Open-v2 evidence identifies a concrete identity-agenda risk for scholarly deduplication. Single-space scientific representation baselines preserve topical relatedness, but their merge threshold can produce high false-merge rates on OpenAlex silver hard negatives. IAD-Risk transformer variants keep high same-work F1 on the held-out Open-v2 test split and show zero observed HNFMR in the agenda-hard-negative stratum, while ordinary FMR=0.001 is still reported for all non-identity rows.

The result interpretation is deliberately narrow. The same IAD-Risk rows still report FMR=0.001, so zero observed HNFMR should be read as no observed false merge in the agenda-hard-negative stratum, not as absence of all non-identity false merges. RoBERTa pair classification is a strong baseline and narrows the gap substantially; its row shows that a supervised pair classifier can already suppress many unsafe merges under the full Open-v2 scope. The IAD-Risk contribution is therefore an auditable relation-role design rather than a broad superiority claim: identity, agenda, and agenda-non-identity signals are exposed before a fixed risk gate decides whether a related pair may merge, block, or defer. The main result is that relation heads plus an explicit risk score provide an auditable mechanism for false-merge suppression under the reported Open-v2 contract.

## 6.22 Result Interpretation Guardrails

The full result interpretation guardrails table is reported in the supplementary material. The Open-v2 table should be read as a mechanism profile with an artifact-release boundary until external artifact release and same-scope prediction files are available. Its Scope type column separates full available Open-v2 scope rows from held-out Open-v2 test scope rows. Scope labels prevent ranking interpretation because the row families do not yet share the same released prediction scope. The intended reading is explicit: representation rows test false-merge exposure, the RoBERTa row is a strong supervised comparator, and IAD-Risk rows test split-held-out risk gating with identity, agenda, and ANI signals.

The zero-HNFMR rows also have a narrow reading. The ordinary FMR=0.001 is still reported for all non-identity rows, so zero observed HNFMR should be read as no observed false merge in the agenda-hard-negative stratum, not as absence of all non-identity false merges. The table is not a claim of broad method superiority, not a same-scope comparative ranking, and not evidence of threshold stability or zero risk.

### 6.23 Manual Validation Boundary

Manual validation is not completed in the current manuscript package. Silver hard negatives are stress-test evidence for identity-agenda false-merge risk, but they are not human-gold non-identity labels. The full manual validation boundary table is reported in the supplementary material. Stronger label-precision claims require a 500–1,000 pair reviewed slice covering hard negatives, high-score false-merge candidates, blocked or deferred pairs, version boundaries, identifier conflicts, and sparse metadata. The protocol requires two independent reviewers who are blind to model scores and merge decisions, plus an adjudication log, agreement report, pair-level notes, and checksums. The paper therefore supports bounded false-merge stress testing under Open-v2, but does not claim complete human validation or that silver rows replace human-gold non-identity labels.

### 6.24 Scope Compatibility of the Open-v2 Table

The Open-v2 table is a scope-bounded evidence table, not a single comparative ranking. The full scope compatibility matrix is reported in the supplementary material. Full pair-scope representation baselines test whether single-space scientific relatedness can over-merge agenda-level hard negatives across the available Open-v2 pairs. Held-out IAD-Risk rows test whether risk-gated transformer variants preserve same-work performance and suppress hard-negative merges under a split-based model-evaluation protocol. The valid comparison is mechanistic: semantic relatedness alone is unsafe as automatic merge evidence, while explicit risk gating can block hard-negative merges in the reported held-out setting.

The stronger ranking path is deliberately left as a future evidence requirement. It requires all row families to be regenerated with the same released prediction scope, threshold-selection logs, checksums, and an independent manual-validation slice. A claim that this stronger comparison has already been completed is outside the current evidence snapshot.

### 6.25 Extended Protocols and Boundaries

Open-v3 and source-heldout protocols are retained as follow-up evaluation paths, not as additional result evidence in the current manuscript package. Their role is to define how source-diverse and domain-diverse validation should be audited before broader claims are made. The manuscript therefore treats Open-v2 as the core mechanism demonstration and reserves Open-v3/source-heldout conclusions for a released artifact package that contains matched prediction scopes, threshold logs, checksums, and manual-validation evidence.

For a source-heldout claim to become admissible, the artifact package must declare train, validation, and held-out source partitions before model fitting; keep source identifiers out of predictive features; report per-source denominators for same-work F1, FMR, and HNFMR; and include command logs, split summaries, prediction checksums, and threshold records for each held-out source. If any gold, proxy, or silver label stratum is missing from the held-out source partition, the result can be reported only as a coverage gap or exploratory diagnostic. Under the current package, source-heldout is therefore a readiness protocol rather than current evidence of source generalization.

## 7 Mechanism and Error Analysis

The central mechanism question is whether false-merge control follows from explicit agenda-non-identity supervision and the derived risk score. The supported Open-v2 comparison gives mechanism-consistent evidence: single-space scientific representations can over-merge agenda-related hard negatives, while IAD-Risk makes agenda-non-identity evidence a direct blocking signal. Within the current evidence, the mechanism reading is triangulated rather than component-causal: representation baselines expose relatedness-driven merge risk, RoBERTa anchors a supervised pair-classification comparator, and IAD-Risk rows show relation-role separation plus risk gating under the fixed operating point.

This mechanism reading does not by itself establish a complete causal ablation. Isolating the effect size of each relation head remains an ablation-artifact requirement. The full mechanism-evidence boundary table is reported in the supplementary material. The main text keeps the mechanism boundary as a prose rule: topical relatedness creates merge risk only as a targeted failure-mode claim; explicit risk gating supports the reported Open-v2 contract and thresholds; component-causality claims require ablations removing risk gating, ANI supervision, and single-space separation; and cluster-level contamination claims require sufficient cannot-link coverage and cluster-level artifact audits.

The full error taxonomy table is reported in the supplementary material. The main text keeps the taxonomy as a prose rule: same task, different contribution pairs, citation-neighborhood neighbors, version or extension boundaries, identifier conflicts, and sparse metadata cases are the main failure modes the framework is designed to expose. This taxonomy is diagnostic rather than a measured error distribution unless a released artifact contains per-category annotations and adjudication logs. The distinction matters because version boundaries, identifier conflicts, and sparse metadata cases require human judgment beyond metadata-derived silver evidence.

The error analysis emphasizes claim discipline. The current evidence supports identity-agenda risk modeling, stratified benchmark construction, and targeted false-merge suppression. It does not support claims that OpenAlex labels are human gold, that the system has broad domain superiority, that component-level ablations have already closed every mechanism question, or that manual annotation has already been completed.

## 8 Limitations

This study has five limitations, and they should be read as scope boundaries for the present evidence package rather than as claims that the false-merge mechanism is resolved in every deployment setting. First, OpenAlex/OpenCitations hard negatives are silver labels, not human gold. They are useful for stress testing but must be reported separately from DeepMatcher gold labels. Second, SciRepEval/SciDocs-style proximity is a proxy for agenda relatedness, not proof of identity or non-identity.

The remaining limitations concern generalization, deployment, and independent validation. Third, source-heldout generalization is not established by the current package and requires declared source partitions, per-source denominators, prediction checksums, and source-level split summaries before making broad claims across scholarly domains. Fourth, pair-level metrics do not by themselves establish cluster-level deployment quality or operational workload reduction; cluster assignments, cannot-link coverage, cluster contamination rate, automatic merge coverage, defer rate, and review-capacity records are required before making cluster-level quality or net-benefit claims. Fifth, an independently reviewed 500–1,000 pair validation slice would provide stronger evidence

about label precision and difficult boundary cases; the supplementary material specifies the sampling, blinding, adjudication, and agreement-reporting protocol required before making stronger human-validation claims.

These limitations define the evidence needed to upgrade the manuscript claims. A stronger label-quality claim requires the reviewed manual-validation slice and adjudication records. A stronger robustness claim requires source-heldout and topic-heldout evaluations generated from the same documented pipeline. A stronger ranking claim requires same-scope prediction files, threshold logs, bootstrap intervals, and checksums for every compared row. A stronger deployment claim requires cluster artifacts that connect pair decisions to final components. Until those artifacts are available, the manuscript should be read as bounded evidence for pair-level false-merge control under the Open-v2 evidence snapshot.

## 9 Threats to Validity

Construct and internal validity depend on whether labels, splits, and thresholds preserve the intended relation semantics. DeepMatcher pairs provide same-work identity evidence, but OpenAlex/OpenCitations hard negatives are silver stress-test pairs rather than manual non-identity gold. Threshold-sensitive systems must select thresholds on validation data and report held-out metrics so that test rows are not used to choose operating points.

External validity and reproducibility depend on the source mix and artifact chain. The Open-v2 evidence supports the identity-agenda mechanism under the reported source mixture, while broader claims require more source-heldout and domain-diverse evaluation. Reproducibility depends on public data availability and stable processing scripts because the repository deliberately avoids redistributing large third-party data files.

The full validity-threats matrix is reported in the supplementary material. The main text keeps the threat model as a prose rule: construct validity is bounded by label strata; internal validity is bounded by threshold and split separation; external validity is bounded by the current source mix; conclusion validity is bounded by the absence of complete causal ablations; reproducibility is bounded by source and artifact availability; and operational validity is bounded by the gap between pair-level decisions and cluster-level deployment. These safeguards do not turn proxy or silver evidence into human-adjudicated truth, and full numeric audit requires L2/L3 artifacts.

## 10 Claim Interpretation Boundary

The full claim-interpretation boundary table is reported in the supplementary material. The main text keeps the boundary as a prose rule: contribution clarity is tied to the IAD-Bench contract, identity-agenda confusion, and HNFMR as a false-merge safety problem; writing reproducibility is limited to code-level checks, fixture rebuilds, schema validation, and artifact-release preparation; experimental strength is limited to the Open-v2 evidence snapshot; evaluation completeness is limited by artifact-backed ablations, threshold grids, and a manual-validation slice; and method design soundness remains bounded by source-heldout validation, topic-heldout checks, and failure-case analysis before broad deployment claims.

The claim-upgrade ladder is therefore explicit. The current manuscript may claim a bounded pair-level mechanism result on Open-v2. A full numerical-audit claim additionally requires same-scope prediction files, threshold logs, source-input manifests, processing-run logs, metric tables, and checksums. A statistical or component-causality claim additionally requires bootstrap intervals, predefined tests, and accepted ablation variants under the same row scope. A deploy-

ment or broad method-ranking claim additionally requires source-heldout or topic-heldout evidence, manual-validation evidence, review-load records, and cluster-level artifacts. Until a higher level is satisfied by released artifacts, the abstract, conclusion, cover letter, and reviewer responses should remain at the lower supported level.

## 11 Conclusion

IAD-Risk addresses a specific failure mode in scholarly work deduplication: semantically related records can be mistaken for the same work when agenda evidence is collapsed into a single identity score. The core idea is to separate identity and agenda evidence before a pair is allowed to become an automatic merge. By adding agenda-non-identity supervision and a false-merge risk gate, the framework turns scholarly deduplication from an unconditional similarity decision into a selective safety decision that can block or defer pairs whose relatedness is more consistent with false-merge risk than work identity.

The Open-v2 evidence supports a conservative pair-level conclusion: identity-agenda disentanglement can provide targeted false-merge suppression under the reported pair scope and operating points. In the evidence snapshot, single-space representation baselines expose hard-negative false-merge risk with HNFMR 0.790–0.999 on the full pair scope, while IAD-Risk transformer variants report same-work F1=0.980 and zero observed HNFMR in the held-out hard-negative stratum, with ordinary FMR still reported separately as 0.001. These rows should be read as scope-bounded mechanism evidence rather than as a same-scope comparative ranking.

IAD-Bench contributes a reproducible benchmark contract by keeping gold identity labels, proxy agenda evidence, and silver hard negatives separate throughout training, evaluation, and claim interpretation. This separation is the main practical lesson of the study: relatedness evidence is useful for retrieval and candidate generation, but it must be assigned an explicit evidence role before it can influence a merge decision.

The takeaway is bounded but actionable. The manuscript does not claim cluster-level deployment quality without cluster artifacts, and it does not claim broad method ranking across scholarly domains. Additional validation should be read as a structured route for upgrading the claim rather than as evidence already present: manual-review slices would test silver-label precision; source-heldout and topic-heldout partitions would test generalization; same-scope baseline predictions, threshold-sensitivity grids, and ablation outputs would test ranking, threshold, and mechanism stability; and cluster-level artifact audits would connect pair decisions to production-style components. Until those materials are released, the contribution remains the auditable pair-level safety mechanism demonstrated on the Open-v2 evidence snapshot.

## Data and Code Availability

The project repository contains the version-controlled source code and CLI entry points, benchmark construction scripts, small public fixtures, fixture tests, schema contracts, data-processing commands, artifact-release guidance, public-release checks, and manuscript sources. The installable package source is under `src/iad_sieve`; `pyproject.toml` declares the package root and exposes `iad-sieve = iad_sieve.cli:main`. The command `python -m iad_sieve.cli --help` verifies command discovery before fixture rebuild or artifact commands are run. The fixture rebuild verifier starts with the same CLI discovery check and rejects missing public-source reconstruction commands before writing fixture outputs.

The executable public-source reconstruction chain runs through `python -m iad_sieve.cli` subcommands: `prepare-deepmatcher`, `prepare-scirepeval-proximity`, `fetch-openalex-works`, `prepare-openalex-weak-labels`, and `build-iad-bench`. This data-processing path is version-controlled in Git and writes schema-bound JSONL and summary outputs rather than raw-data copies. Raw third-party source files, full prediction files, model checkpoints, and derived evaluation artifacts remain outside Git unless released through an external artifact package. Raw third-party data are not redistributed; the repository avoids redistributing raw data or full experiment outputs in Git because the original sources have their own access conditions and licenses. External releases should provide derived tables, predictions, logs, manifests, and checksums rather than raw provider files unless original provider terms explicitly allow redistribution; commit identifiers and each source entry’s license boundary must remain recorded.

The full data and code availability boundary table is reported in the supplementary material. The boundary separates L0/L1 code-level reproduction from L2/L3 result-level audit: reviewers with only the Git repository can run fixture rebuild validation via `python manuscript/scripts/verify_fixture_rebuild.py`, the public-release audit via `python scripts/check_public_release.py`, and the manuscript validation scripts. These checks verify executable contracts, fixture behavior, public-release boundaries, and manuscript builds, but they do not reproduce the Open-v2 numerical table, model predictions, threshold choices, or bootstrap intervals. Full numerical reproduction requires public-source rebuilds or released artifacts with `source_input_manifest` and `processing_run_log` records for public input provenance, commands, software environments, seeds, output paths, exit status, and checksums.

An external artifact package should be treated as result evidence only after artifact-level and manuscript-package validation pass. Before copying result files, `manuscript/scripts/populate_artifact_release.py` with `--artifact-dir /path/to/release`, `--source-dir /path/to/source-artifacts`, and `--preflight-only` checks that the source artifact directory contains the files required by the release manifest without writing release files; the source preflight can detect missing tables, predictions, reports, configurations, or logs early, but it is not itself result evidence. The release is checked with `manuscript/scripts/validate_artifact_release.py --artifact-dir /path/to/release`, and the final upload package is checked with `manuscript/scripts/validate_submission_package.py --final-upload --artifact-dir /path/to/release`. These checks bind the artifact manifest, submission metadata, source-control commit, and final package to the same source commit; if validation fails, the artifact package should not be used to support the Open-v2 numerical table or any stronger claim.

## Ethics Statement

This study uses public scholarly metadata and benchmark-style record linkage data. It does not involve human participants, clinical records, private user behavior, or animal experiments. Any future manual validation should be limited to scholarly record relation labels and should avoid collecting sensitive personal information.

## Competing Interests

This anonymous review file does not assert a finalized competing-interest declaration. Before final upload, the listed authors must confirm the competing-interest status and synchronize the final statement with `submission_metadata.yml` and the live submission system.

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